Behavioral Clustering of Jungian Types: Linking the 4F Trauma Modes, Big-Five Traits, and Cognitive Functions

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Abstract

This study shows that each Jungian type predominantly operates in one of four survival modes—Freeze, Fight, Fawn or Flight—producing characteristic Big-Five profiles. We test this claim by clustering 2,916 participants (mean $n_{\text{type}} = 182.3$) on their Conscientiousness, Extraversion, Agreeableness and Neuroticism vectors (Openness excluded), evaluating whether emergent clusters align with TRPI's 4F/Big-Five mapping and the underlying Jungian functions. K-means ($k = 4$) with k-means++ seeding and a reproducible RNG produced cluster assignments. A permutation ^{2} test (50 000 iterations) assessed whether cluster memberships were independent of both dominant trait and Jungian type. Four distinct centroids emerged: high C (Freeze), high E (Fight), high A (Fawn) and high N (Flight). Chi-square values exceeded the 99.9th percentile of the permutation distribution ($p_{\text{Trait}} = .00002$; $p_{\text{Type}} = .00002$), indicating non-random alignment. Behavioural clustering corroborates TRPI's linkage of survival modes, Big-Five traits, and Jungian function pairings, thus supporting the existence of function-driven personality states beyond static type labels.

Keywords: Cognitive functions, 4F trauma responses, Big Five, k-means clustering, Jung, TRPI

1. Introduction

Jungian typology offers cognitive functions (e.g., *Se*, *Ni*) but lacks a data-driven behavioural taxonomy. Recent work [3] demonstrated that Jungian types show reliable Big-Five signatures. Building on that empirical base, the present study asks:

> *Do natural trait clusters map onto the 4F's survival modes and, by extension, onto the dominant function pairings predicted for each type?*

2. Method

2.1. Participants and Measures

We analyzed N=2,916 TRPI profiles collected via traitindicator.com. Each record contains Jungian type and 26-item Big-Five responses (Cronbach $\alpha = .70$–$.97$).

2.2. Trait Vector Preparation

Openness indexes the *perceiving* side of cognition—specifically the Sensing/Intuition axis—whereas the four survival modes (Freeze, Fight, Fawn, Flight) arise from *judging-function* polarity (Te/Fi vs. Fe/Ti). Because the present study tests how the 4F modes partition behavior, only trait dimensions linked to judging functions (C, E, A, N) are required; including Openness would add perceiving variance that is theoretically orthogonal to the hypothesis. The resulting 4-D vector $\mathbf{x} = [C, E, A, N]$ was z-standardized.

2.3. Clustering and Validation

K-means ($k = 4$) was applied with seeded RNG for reproducibility. Assignment stability was checked across 30 bootstrap replicates (mean ARI = .96). Independence of clusters from dominant trait and Jungian type was tested via permutation ^{2} (50,000 shuffles).

3. Results

3.1. Emergent Centroids

Table 1: Cluster centroids in z-units and theoretical labels.

Cluster	C	E	A	N	Label
0	1.34	–0.15	–0.12	–0.08	Freeze
1	–0.05	1.21	0.03	–0.14	Fight
2	–0.11	–0.04	1.28	–0.09	Fawn
3	–0.17	–0.18	–0.13	1.37	Flight

3.2. Cluster × Dominant Trait

Observed $\chi^2(9) = 612.1$, $p_{\text{perm}} = .00002$. Conscientiousness-dominant profiles concentrated in Cluster 0, Extraversion in Cluster 1, Agreeableness in Cluster 2, and Neuroticism in Cluster 3, matching TRPI expectations.

3.3. Cluster × Jungian Type

Jungian type distributions mirrored their hypothesised survival modes (e.g., 76% of INTJs in Freeze; 68% of ESFPs in Flight).

4. Limitations

1. **Cross-sectional snapshot.** Longitudinal stability and mode transitions were not assessed.
2. **Algorithmic assumptions.** K-means enforces spherical clusters; alternative methods might reveal sub-modes.
3. **Openness exclusion.** While theoretically justified, omitting Openness prevents assessing perceiving-function moderation.
4. **WEIRD sample.** Online, English-speaking volunteers limit cultural generalizability.

5. Conclusion

Behavioral clustering of four Big-Five traits reproduces the 4F's Freeze, Fight, Fawn, and Flight modes and aligns those clusters with Jungian function pairings. The findings suggest that survival modes are observable data-driven states linking traits to dynamic cognitive processes.

References

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